

CARGOMATE
SMART ROBOTIC INVENTORY SYSTEM WITH CLOUD INTEGRATION

25-26J-032

Project Proposal Report

Perera R.A.G

B.Sc. (Hons) Degree in Information Technology Specialized in Computer Systems
& Network Engineering

Department of Information Technology
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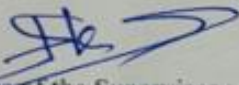
Declaration of the Candidate & Supervisor

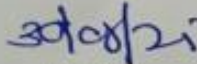
Declaration of the Candidate & Supervisor

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any previously published or written by another person material, except where the acknowledgement is made in the text.

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The above candidate are carrying out research for the undergraduate Dissertation under my supervision.


Signature of the Supervisor :


Date:

Abstract

The rapid growth of smart warehousing [1] has shown the need for efficient stock management, proactive replenishment, and strong network performance monitoring. This project component, titled Smart Warehouse Inventory System with Cloud Integration, focuses on integrating Internet of Things (IoT) sensor data with cloud-based systems to ensure resilience and efficiency. The system will collect data from sensors, which will be implemented by team members, and store it in Firebase. A dashboard with Role-Based Access Control (RBAC) will be developed to visualize stock levels securely. Additionally, Grafana will be integrated for network performance monitoring with custom Firebase metrics. A multi-channel alerting system will notify warehouse robots to start stock refilling when levels drop below certain thresholds. At the same time, email alerts will ensure human intervention for resilience. A digital twin of the warehouse will be created to simulate and optimize replenishment scenarios. The expected outcomes include better visibility of stock levels, proactive decision-making, improved network reliability, and the integration of a digital twin for predictive simulation.

Keywords: Warehouse Management, IoT, Firebase, Grafana, Digital Twin, Role-Based Access Control

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Chapter 1 INTRODUCTION

1.1 Background

Warehouse management is a key part of modern supply chains through the provision of effective storage, retrieval, and distribution of goods. With expanding global commerce and demands for faster delivery mounting, efficiency in warehouse operations has become a necessity for business success. Historical storage facilities that rely on manual operations for inventory tracking and reordering are hampered by a number of problems like inventory lag in monitoring, human errors in data collection, and inefficiency in responding to stock-out conditions. These breakdowns have a tendency to lead to disruption of activities, increased costs, and reduced customer satisfaction.

Over the last couple of years, the use of Industry 4.0 technologies revolutionized warehouse management. IoT sensors enable real-time monitoring of inventory quantities, weather conditions, and equipment health. Such data, when stored in cloud platforms such as Firebase, can be retrieved quickly, securely, and at scale. Cloud storage does not only increase efficiency but also interstaff collaboration and management through a centralized console accessible anywhere.

Another significant advance is the development of real-time monitoring dashboard systems. Dashboards consolidate warehouse data into graphical, interactive reports that facilitate data-driven decisions from decision-makers. To enable safe use, modern dashboards employ Role-Based Access Control (RBAC), which allocates access permissions based on roles. For example, an administrator can modify stock quantities, while a warehouse operator can read and modify item quantities. This prevents abuse and enhances the overall safety of the system.

Besides tracking inventory, warehouse operation resilience is also a requirement. Early detection of stock depletions in a resilient system can trigger automatic response in the form of notifications that not only reach administrators but also autonomous robots filling shelves. This reduces dependency on human intervention and enables constant supply of stock. For added resilience, multi-modal notifications such as auto-email notifications are incorporated that enable warehouse managers to respond quickly to potential disruptions.

Also, as more and more dependency is laid on interdependent systems, monitoring of network performance has become a critical component of warehousing. Tools such as Grafana is used to

monitor the network health, latency, and connectivity among IoT devices, dashboards, and robots in real time. Custom Firebase metrics added ensure that not only the infrastructure level but even application-level monitoring is covered, offering deeper insights into performance.

Another innovation comes with the addition of the digital twin. The digital twin is a virtual representation of the warehousing space that mirrors real-world conditions. With integration of real-time sensors, the digital twin can predict stock flows, anticipate shortages, and analyze operational changes before implementing them in the physical space. This ability to forecast aids in improved decision-making and increased warehouse efficiency.

1.2 Literature Survey

Warehouse management has been a big research field within logistics and supply chain management, where huge focus has been laid on automation, optimization, and digitization. A variety of methods have been proposed by academics and practitioners to make warehouse operations efficient, ranging from process redesign to adoption of smart technologies. The review below presents significant areas of related work to this project.

1.2.1 IoT in warehouse management

The Internet of Things (IoT) has significantly influenced today's warehouse systems. Researchers have indicated that IoT sensors can record real-time data on stock levels, shelf life, and ambient parameters such as temperature and humidity. Literature also maintains that IoT-based systems minimize human errors and improve decision-making by offering accurate and timely data. The majority of these systems, however, face delays in terms of scalability, data protection, and integration with robot systems.

1.2.2 Role-Based Access Control (RBAC) Dashboards

RBAC has been a key security practice for enterprise systems. It ensures that different users, such as admins, managers, and operators, only access authorized resources. Research in e-governance and financial dashboards shows that RBAC improves data confidentiality and accountability. However, studies indicate that in warehouse IoT contexts, dashboards often show flat access models. This means all users see the same data. Such setups can leave systems open to misuse and accidental configuration errors. By integrating RBAC into a warehouse dashboard, we can separate duties. For example, only supervisors can approve replenishments, while staff can only view stock levels. This approach helps fill the existing gap.

1.2.3 Network Performance Monitoring with Grafana

Robust network performance is paramount in a smart warehouse, where robot interaction, Firebase cloud storage, sensor data, and dashboards all depend on secure integration. Disruptions or latencies in these links can have an impact on real-time decision-making as well as stock restocking. Grafana provides an adaptable solution with real-time dashboards and visualizations that allow monitoring of system health and data streams. In contrast to generic monitoring of only hardware resources, Grafana may be configured with custom metrics relevant to warehouse activities, such as the delay between sensor refresh and Firebase storage, dashboard response time, and delay in resupplying a robot. Such metrics give stakeholders meaningful insights into operational productivity rather than infrastructure utilization. Also, the built-in alerting facility of Grafana ensures reliability by alerting administrators whenever the performance departs from standard limits, thereby enabling timely intervention. Overall, Grafana is not only responsible for monitoring network and application reliability but also consolidates warehouse automation by ensuring round-the-clock communication between all elements.

1.2.4 Multi-Channel Alerts (Robot Triggers + Email Notifications)

Warehouse resilience research shows that weakness is possible if a single notification channel (SMS or dashboards) is utilized and either one of them is down. By combining robot fill orders with automated email reminders to managers, the project gains a double resistance mechanism.

1.2.5 Digital Twin Integration

The digital twin concept—solving physical systems with an imitation model—has been a hotspot in many industries, particularly manufacturing and logistics. Digital twins assist in simulating operations, predicting bottlenecks. Digital twin usage in warehouses has so far been largely restricted to big industries, leaving small warehouses with cost and complexity challenges.

1.2.5 Robotic Integration in Warehousing

Autonomous robots increasingly utilized in warehouses to execute repetitive tasks such as picking, sorting, and restocking. This shows that there is a need for systems with the ability to automatically trigger refill operations through real-time alerting from stock monitoring systems to robots.

1.3 Research Gap Table

While today's WMS are focused on automation, they don't focus on IoT real-time data integration, network performance monitoring, resilience notifications, and predictive simulations using digital twins. This deficiency undermines proactive decision-making and system reliability.

Table 1.1: Gap Table

Features	Current Research	Identified Gap	Proposed Solution
Dashboard & Role-Based Access (RBAC)	Current DT research emphasizes visualization via BIM-based dashboards, but limited focus on access control and user roles [2]	Insufficient security and user-specific views for warehouse stakeholders.	Implement RBAC in Firebase Auth, ensuring admins/operators/viewers have different dashboard privileges.
Network Performance Monitoring	Research focuses mostly on warehouse operations but neglects the underlying IT/network reliability [3]	Gap in integrating warehouse WMS with network performance monitoring tools.	Integrate Grafana to monitor system and network performance, ensuring stable connectivity for WMS and robots.
Multi-channel Alert System	Basic ledger recording with manual reporting processes [4]	Limited alert mechanisms and no redundant notification systems	Implement dual alert system: robot notifications for immediate action and email alerts to staff for resilience and backup
Digital Twin Integration	Digital twin concepts mentioned in Industry 4.0 context but limited practical implementation in warehouse	Lack of integrated digital twin solutions for real-time warehouse state monitoring and predictive analytics	Integrate digital twin technology to create virtual warehouse replica that mirrors real-time conditions and enables predictive maintenance and optimization

	management systems [5]		
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1.4 Research Problem

Modern warehouses are evolving into smart cyber–physical environments with IoT sensors, cloud infrastructure, and robots collaborating to manage inventory. However, despite advancements in technology, there are a few issues still unresolved that hinder warehouse efficiency and resilience

- **Lack of Real-Time Visibility:** There is no batch or periodic update offered by conventional Warehouse Management Systems (WMS). This delay deprives timely refilling and causes stockouts or overstocking. Although IoT sensors can capture data in real-time, various systems lack the proper database infrastructure to synchronize and present such updates to various stakeholders.
- **Insecure and Uniform Dashboards:** The majority of dashboards offer a flat model of access, where everybody views the same thing. This introduces security threats (e.g., unauthorized staff initiating replenishment orders) and usability issues (e.g., cluttered dashboards with irrelevant data). The absence of Role-Based Access Control (RBAC) results in business inefficiencies as well as data privacy concerns.
- **Neglected Network Reliability:** Fact confirms network performance to be of greatest importance in robotized warehouses, where robots depend on low-latency messaging with cloud platforms. Most warehouse projects, however, address merely the sensor layer, disregarding ongoing monitoring of network health. Network outages can cause robot refills to be held up, stock shortages a foregone conclusion.
- **Single-Channel Notification Limitations:** Current systems only provide one type of notification (e.g., dashboard messages or SMS). Should the primary channel fail, critical stock replenishment messages may never reach responsible staff. This offers a single point of failure, impacting resilience during times of high demand or fault scenarios.

Chapter 2 OBJECTIVES

2.1 Main Objective

The main objective of this project module is to build and implement an intelligent warehouse monitoring system with real-time visibility, secure access control, strong alerting mechanisms, and predictive simulation for round-the-clock stock replenishment and operational efficiency.

2.2 Specific Objective

- **Collect and store warehouse sensor data in Firebase:** Establish a real-time database schema in Firebase to collect streams of data from IoT sensors deployed in the warehouse, ensure stock levels are updated from time to time and accessible for downstream activities (dashboards, monitoring, alerts, and digital twin).
- **Develop a dashboard with Role-Based Access Control (RBAC):** Create an easy-to-use dashboard that graphically displays stock levels and system status, Implement RBAC in a way that different roles (e.g., Admin, Supervisor, Operator) are assigned tailored access rights—improving security and usability, Limit sensitive activities (e.g., replenishment authorizations) to authorized personnel only.
- **Integrate Grafana for network performance monitoring:** Deploy Grafana for advanced data visualization, Add the monitoring to include Firebase custom metrics, correlating stock-level events with network performance, provide reliable robot and dashboard communication by actively monitoring network delays or shutdowns.
- **Implement a multi-channel notification system (robot trigger + email alerts):** Develop an automated notification system that instantly calls on warehouse robots for replenishment when levels are attained, Build redundancy by sending email alerts to supervisors, ensuring human awareness in case robot commands fail or network issues arise, This two-channel method enhances resilience and minimizes the likelihood of missed replenishment measures.
- **Integrate a digital twin for simulation and predictive optimization:** Develop a digital twin that mirrors warehouse operations in real time, utilize it for what-if analysis (for example, network downtime, robot delay, demand surge), Streamline replenishment strategies and test out alerting mechanisms without affecting live operations

Chapter 3 Methodology

3.1 System Diagram

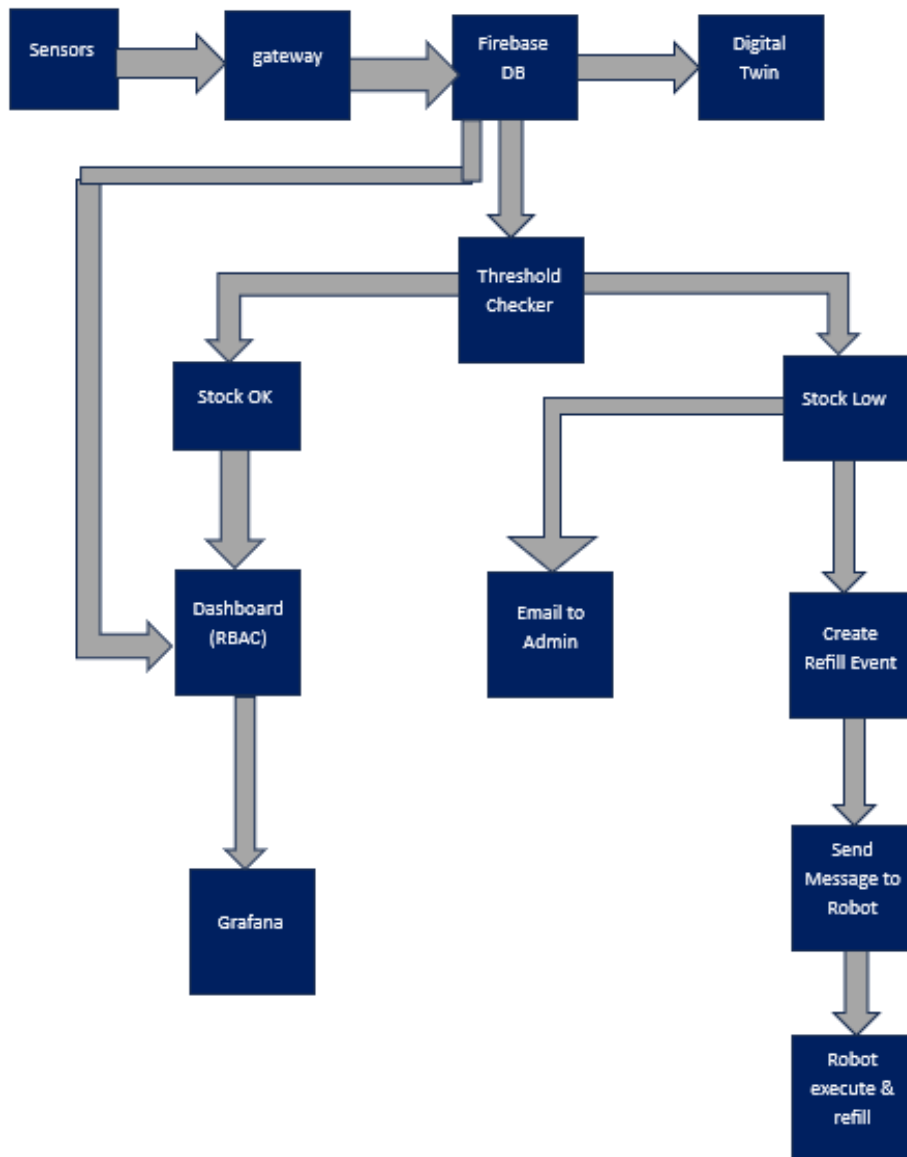


Figure 3.1: System Diagram

1.Sensors → Ingestion → Firebase

- Stock levels get scanned by sensors.
- A small app receives that data.
- It saves all to Firebase (your cloud database).

2.Firebase is read by Digital Twin

- A twin service copies the same data so you can visualize the warehouse.

3.Threshold Check (stock in / stock out).

A Cloud Function checks each item:

- If $qty \geq \text{min Qty}$ → status = stock in
- If $qty < \text{min Qty}$ → status = stock out

4.If status = stock in

- Dashboard (RBAC) shows normal stock.
- System also triggers "STOCK IN" message to navigation module.
- Admin/Staff users receive live updates by user role.

5.If status = stock out

- Trigger a Refill Event in Firebase.
- Send a "REFILL" message to Robot.

6.Admin Email (Resilience)

When we observe stock out, the system sends an alert to the admin via email.

3.2 Component diagram

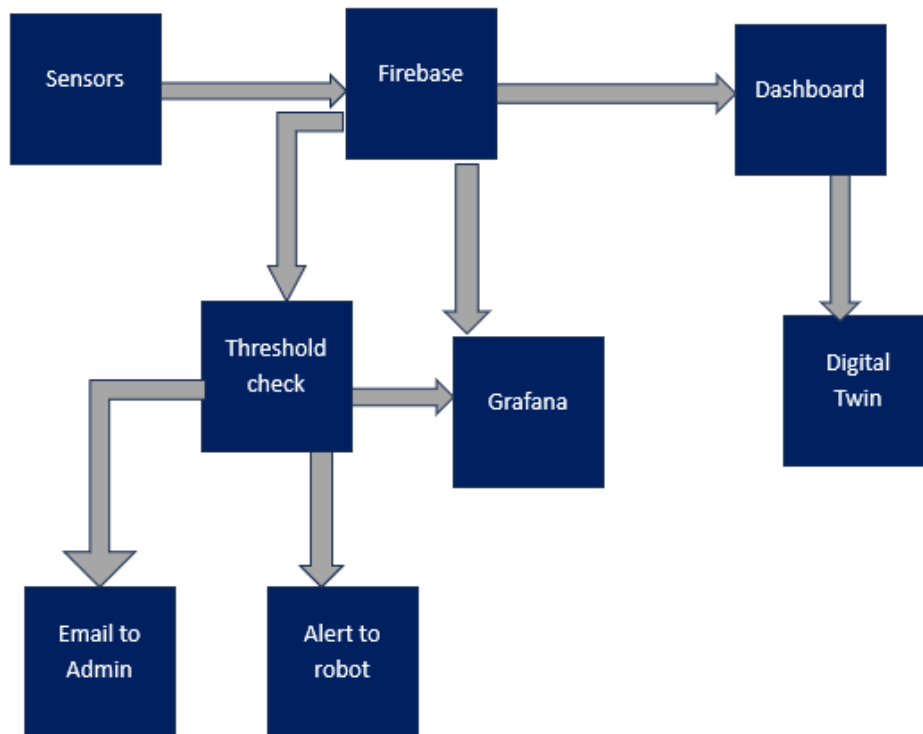


Figure 3.2: Component Diagram

- Ingestion Gateway -Secure entry point for sensor data. Writes the data to Firebase.
- Firebase Auth (RBAC)-Authentication + role-based access. (Roles: Admin, Staff, Robot, Monitor).
- Dashboard -Reads from Firebase (shows inventory, alerts, events timeline. RBAC limits what each role can see/do).

- Threshold-Triggered when items qty changes.

If $qty < \min Qty \Rightarrow$ mark stock out, create refill event.

If $qty \geq \min Qty \Rightarrow$ mark stock in, close events, notify navigation.

- Email Alert Service-Sends email on stock out.
- Navigation Service -Receives “STOCK IN” / location updates after refill to coordinate path planning.
- Digital Twin Service-Mirrors Firebase state for visualization; optional “what-if” playback.
- Metrics Exporter -Extracts app metrics (latencies, counts, error rates) and pushes them to a Grafana-readable source.

How they work together

1. Data & Display
Sensors $\rightarrow$ Ingestion Gateway $\rightarrow$ Firebase DB $\rightarrow$ Dashboard (RBAC) + Digital Twin.
2. Stock-Out $\rightarrow$ Robot Refill
Firebase $\rightarrow$ sees $qty < \min Qty \rightarrow$ create refill event $\rightarrow$ sends command $\rightarrow$ Robot refills $\rightarrow$ ACK $\rightarrow$ Firebase $\rightarrow$ status flips to stock in $\rightarrow$ Dashboard updates $\rightarrow$ Navigation gets “stock in” message $\rightarrow$
3. Monitoring
Grafana do the monitoring part.

Chapter 4 PROJECT REQUIREMENTS

4.1 Functional requirements

- Functional Requirement 01 – Sensor ingestion: Read readings from sensors in warehouse via Gateway and write to Firebase.
- Functional Requirement 02– Inventory model: Save items, sensors, events, alerts
- Functional Requirement 03 – Threshold: On item change, compute stock in / stock out.
- Functional Requirement 04– Event creation: On stock out, create a refill Needed
- Functional Requirement 05 – Robot command: Publish refill message to robot

4.2 User Requirements

01.Admin

- Log in safely, view all items & events.
- Manage users & roles

02.Warehouse Staff

- View inventory & alerts (no config privileges).

03.Monitor

- View Grafana dashboards & alerts

04.Robot

- Receive commands (stock in and out)

05.Navigation System

- Receive "stock in" message for successful refill.

4.3 System Requirements

01.Cloud

- Firebase project (Firebase Auth, Cloud Functions)
- Dashboard web app
- Email provider integrated via Function.
- Grafana server + Metrics Exporter service.
- Digital Twin service/app subscribing to Firebase.

02.Hardware/Network

- Get data from sensors.

03.Security

- RBAC access controlling.

4.4 Use Cases

Use Case01

- Write Sensor Reading
- Actors: Sensor, Gateway, Firebase
- Main flow: Gateway validates → writes to Firebase

Use Case02

- Stock Threshold Check (stock in/out)
- Actors: Cloud Function
- Flow: If qty < min Qty → status=stock out & fire refill event; else status=stock in

Use Case03

- Robot Refill Command
- Actors: Cloud Function, Robot
- Flow: Send command to refill the stock

Use Case04

- Admin Email Alerts
- Actors: Email service, Admin
- Flow: Build & send email

Use Case05

- RBAC Dashboard
- Actors: Admin/Staff
- Flow: Login with Firebase Auth, Admin can modify min Qty

Use Case06

- Grafana Monitoring
- Actors: Metrics Exporter, Grafana, Monitor
- Flow: Export metrics → Grafana panels

Use Case06

- Digital Twin
- Actors: Twin Service
- Flow: Render locations & stock states

4.5 Test Cases

Test Case 01

- Sensor → Firebase write
- When sent to Gateway

Test Case 02

- Threshold sets stock out
- Given item with qty=5, min Qty=10

- Then status=stock out and refill event triggered.

Test Case 03

- Threshold sets stock in
- Given status=stock out
- When qty updated to $\geq$ min Qty
- Then status=stock in.

Test Case 04

- Low stock alert
- When the stock level goes down send an email to the admin for the resilience.

Test Case 05

- RBAC access: Admin can modify min Qty
- Given Staff login
- When attempting to modify min Qty
- Then access denied by Security Rules.

Test Case 06

- RBAC: Admin can modify min Qty
- Given Admin login
- When modifying min Qty
- Then modification takes effect and is audited.

Test Case 07

- Digital Twin reflects stock out
- When the stock level goes down it shows stock out
- When twin refreshes
- IT shows stock out

Test Case 05

- Security rules
- Given unauthenticated request
- When reading items
- Then denied.

Chapter 5 GANTT CHART

Table 5.1: Gantt Chart

Cloud Integration		2025					2026						
Milestone	Month	8	9	10	11	12	1	2	3	4	5	6	7
Research papers collect, Requirements planning													
Data contracts with sensors													
API gateway													
Cloud functions validation													
Fire store manages													
Dashboard													
RABC integrate													
Grafana dashboards													
Custom metrices from firebase													
Alert to robot													
Alert as email													
Digital twin modelling													
End to end testing													
Documentation and finalization													

Chapter 6 WORK BREAKDOWN

Phase 1 – Research and data collection

Phase 2 - Database Setup.

Phase 2 – Dashboard Development (RBAC).

Phase 3 – Grafana Network Monitoring.

Phase 4 – Notification System (Robot and Email).

Phase 5 – Digital Twin Integration.

Phase 6 – Testing, Validation, Documentation.

6.1 Work Breakdown Chart

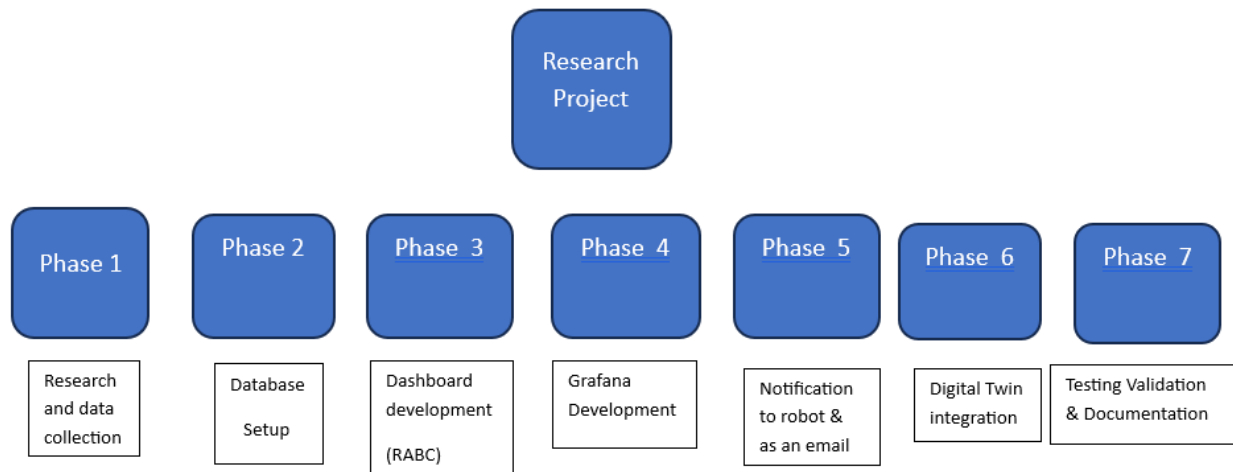


Figure 6.1: Work Breakdown Chart

Chapter 7 DESCRIPTION OF PERSONNEL AND FACILITIES

Facilitators

Mr. Uditha Dharmakeerthi - Sri Lanka Institute of Information Technology (SLIIT)

Prof. Anuradha Jayakody- Sri Lanka Institute of Information Technology (SLIIT)

Chapter 8 DESCRIPTION OF BUDGET

Table 8.1: Budget

component	cost
Internet cost	Rs.3000.00
For the tools	Rs.4000.00
Miscellaneous Expenses	Rs.1000.00
Grafana setup	Rs.2500.00
Firebase cloud plan	Rs.4000.00

Chapter 9 COMMERCIALIZATION

Our commercialization plan is a cloud-first SaaS for small and mid-size warehouses that need faster, more reliable refills. We offer a real-time stock dashboard with RBAC, resilient alerts that trigger robots and email supervisors, network-to-business correlation in Grafana, and a digital twin for “what-if” planning—delivered with quick setup and low upfront cost.

9.1 Target Market

- Pharmaceutical & medical supply warehouses.
- E-commerce fulfillment centers.
- Retail distribution centers & store backrooms.
- Manufacturing plants.
- Postal/courier parcel hubs.

Chapter 10 REFERENCES

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